DIGITAL VERIFICATION

VeriChip ALU and state-machine verification

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I built a self-checking SystemVerilog testbench using directed tests, constrained-random stimulus and assertions. The examples below show how the test plan defines an expected result for each input.

Directed ALU case / SWAP

Register	Before command	Expected after SWAP
Left	0x0301	0x0D1B
Right	0x0D1B	0x0301

Two distinct operands make a partial swap visible. The checker must observe both registers to distinguish a complete exchange from a one-sided write.

State-machine case / Arithmetic overflow

Starting in Normal, the plan calls for an overflowing add or subtract. The expected behavior is a transition to Error with the specified INT1 response.

Once in Error, the next checks exercise register access: reads should remain enabled and writes should be blocked.

Verification tools: SystemVerilog, VCS and Verdi. The examples show planned checks and expected values.

Digital Verification and Testing: ALU test plan (with Akshith Alwala), p. 6; state-machine test plan, p. 3.